Little Red Dot J0923+4157 Glows in Radio

Archival radio data reveal a compact, variable counterpart to a local little-red-dot galaxy.

arXiv:2610.17559 — radio counterpart of the local little-red-dot

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Background

Little red dots are compact, red, high-redshift galaxies whose nature is debated. A rare local example, J092359+415736, lets astronomers study one up close.

Little red dots are mostly known at high redshift; radio emission from a local analogue was unexplored.

What this paper is selling

The authors stack ten archival radio epochs to detect a compact counterpart at the Gaia position. Its radio behaviour and SED help test whether little red dots host AGN-like activity.

Key results to remember

- A compact radio source is detected at the Gaia DR3 position across multiple epochs.
- Radio flux densities vary between epochs, hinting at an active, compact engine.
- Two-component blackbody fit gives cool temperatures of 4583 K and 2086 K.

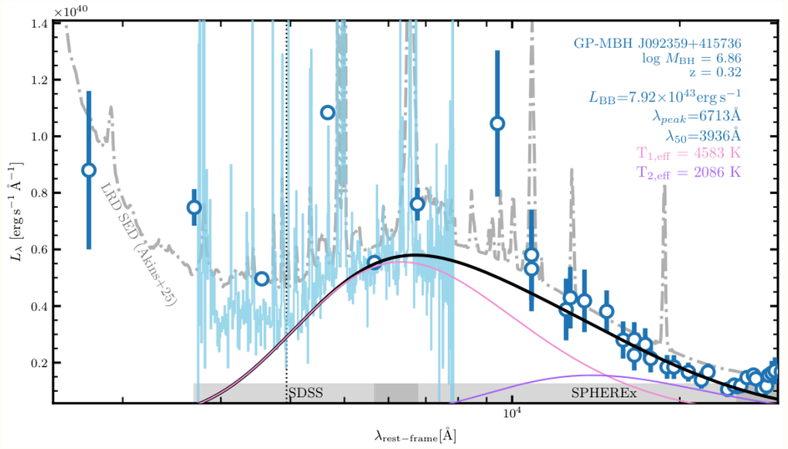


Figure 2 | Two-component blackbody fit to the UV–near-IR SED, placing J092359+415736 among gas-reddened AGN and little-red-dot populations.

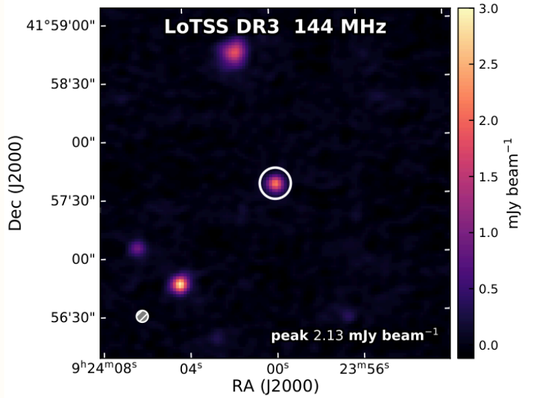


Figure 4 | LoTSS 144 MHz image shows a 42-sigma compact radio source inside the Gaia error circle, confirming a radio counterpart.